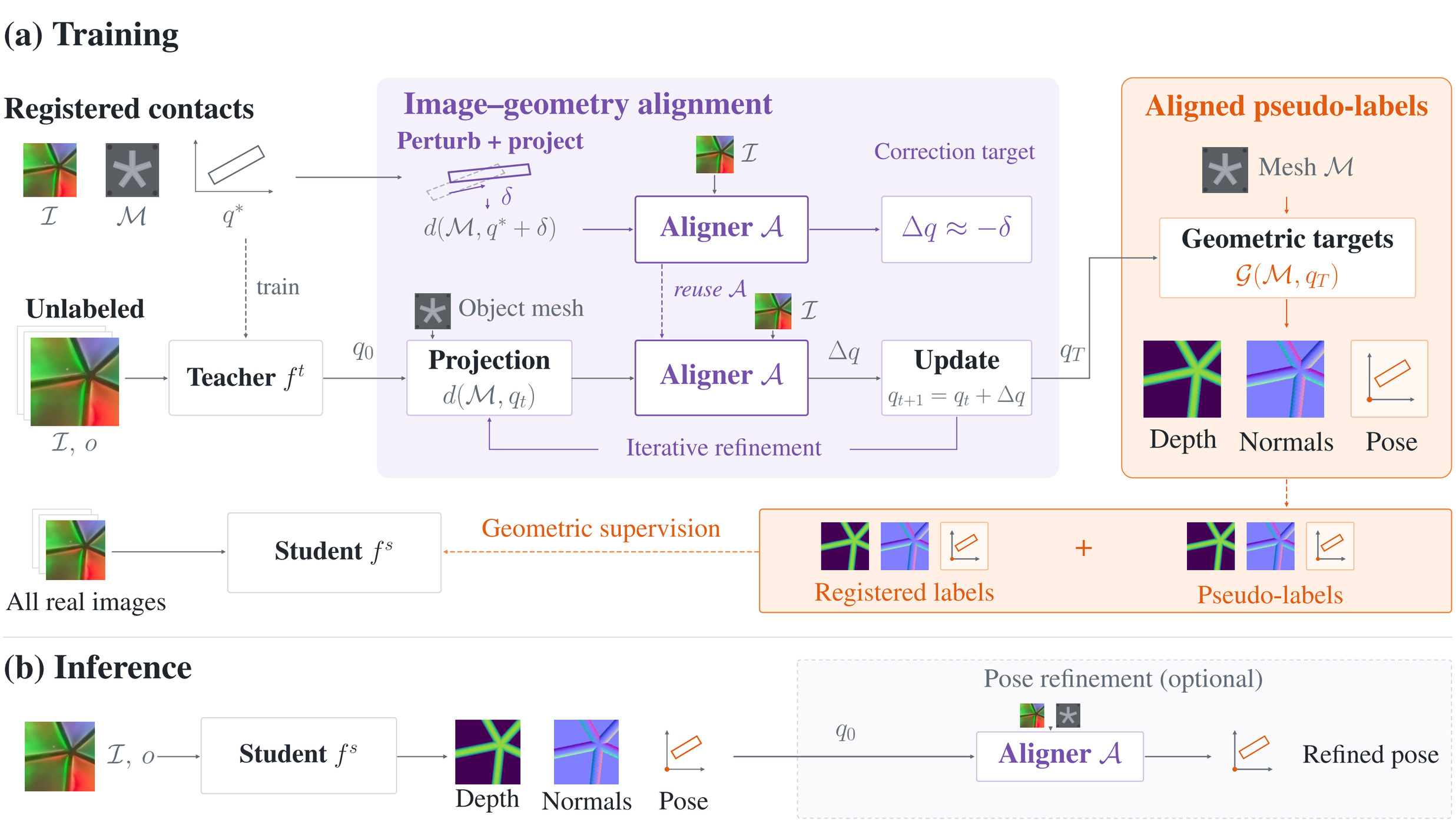

(a) Training
Registered contacts
$\mathcal{I}$
$\mathcal{M}$
q^*
train
Unlabeled
$\mathcal{I}$, o
Teacher f^t
q_0
Image–geometry alignment
Perturb + project
δ
$d(\mathcal{M}, q^* + \delta)$
$\mathcal{I}$
Aligner $\mathcal{A}$
Correction target
$\Delta q \approx -\delta$
reuse $\mathcal{A}$
Object mesh
$\mathcal{I}$
Projection
$d(\mathcal{M}, q_t)$
Aligner $\mathcal{A}$
Δq
Update
$q_{t+1} = q_t + \Delta q$
Iterative refinement
q_T
Aligned pseudo-labels
Mesh $\mathcal{M}$
Geometric targets
$\mathcal{G}(\mathcal{M}, q_T)$
Depth
Normals
Pose
All real images
Student f^s
Geometric supervision
Registered labels
+
Pseudo-labels
(b) Inference
$\mathcal{I}$, o
Student f^s
Depth
Normals
Pose
Pose refinement (optional)
q_0
Aligner $\mathcal{A}$
Refined pose